Capstone Project: Problem Statement and Success Metrics

Professional Problem Statement

> **Problem:** Small clinics and healthcare providers in low-resource settings often lack the specialized tools and personnel required for the **early and accurate identification of high-risk pregnancies**, leading to delayed interventions and contributing to preventable maternal and infant morbidity and mortality. The current reliance on manual assessment and limited data analysis capabilities hinders proactive care management.**Solution:** To address this, we will develop a robust, AI-driven predictive model integrated into a simple web application. This system will utilize routine physiological data (e.g., blood pressure, heart rate, blood sugar) to classify pregnant patients into **Low, Mid, or High-Risk** categories.**Goal:** The primary goal of this capstone project is to demonstrate the feasibility and clinical utility of machine learning in providing an accessible, data-driven decision support tool that empowers small clinics to prioritize care, allocate resources effectively, and ultimately improve maternal health outcomes through timely intervention.

Key Success Metrics for the Data Scientist

As the Data Scientist, your success will be measured not just by the model's overall accuracy, but by its performance on the most critical class:

Metric	Description	Target	Rationale
Recall for 'High Risk' Class	The percentage of actual High-Risk patients that the model correctly identifies.	**> 85%**	**Most Critical Metric.** Minimizing **False Negatives** (missing a high-risk patient) is paramount for patient safety and clinical utility.
Macro-Averaged F1-Score	The harmonic mean of Precision and Recall, calculated independently for each risk class (Low, Mid, High) and then averaged.	**> 80%**	Provides a balanced view of the model's performance across all three classes, especially the challenging Mid-Risk category.

Model Latency	The time it takes for the model API to return a prediction after receiving input data.	**< 500 milliseconds**	Ensures the prediction is near-instantaneous, making the web application responsive and useful for real-time clinical decision-making.
Model Interpretability	The ability to explain *why* the model made a specific risk prediction (e.g., feature importance).	**High**	Essential for clinical trust and adoption. The model should be a decision *support* tool, not a black box.